

Data Structures

Fall 2025

4. Complexity Analysis II

Array Searching

Array Operations

- **Insertion**

- Operation of **adding** another element to an array
- How many steps in terms of **n** (number of elements in array)?
 - At the end
 - In the middle
 - In the beginning
- **n steps** at **maximum** (move items to insert at given location)

- **Deletion**

- Operation of **removing** one of the elements from an array
- How many steps in terms of **n** (number of elements in array)?
 - At the end
 - In the middle
 - In the beginning
- **n steps** at **maximum** (move items back to take place of deleted item)

Array Operations: Search Algorithms

- Operation of **locating a specific data** item in an array
 - Successful: If location of the searched data is found
 - Unsuccessful: Otherwise
- **Complexity** (or **efficiency**) of a search algorithm
 - **Number of comparisons $f(n)$** required to locate data within array
 - **n** is the **number of elements** within array
- Two algorithms for searching in arrays
 - Linear search (or sequential search)
 - Binary search

Linear Search

- Very intuitive and simple algorithm

Algorithm works as follows:

- Starts from the first element of the array
- Uses a loop to sequentially step through an array
- Compares each element with the data item being searched
- Stops when data item is found or end of array is reached

Linear Search Algorithm

```
// numElems - maximum number of elements in the array
// value     - integer data (item) to be searched
// position  - array subscript that holds value (if success)
//           - -1 if value not found

int searchList(int list[], int numElems, int value)
{
    int index = 0;        // Used as a subscript to search array
    int position = -1;    // To record position of search value
    bool found = false;   // Flag to indicate if the value was found
    while (index < numElems && !found)
    {
        if (list[index] == value) {
            found = true;
            position = index;
        }
        index++;
    }
    return position;
}
```

Calling Function searchList

```
#include <iostream.h>
```

```
// Function prototype
```

```
int searchList(int [], int, int);
```

```
const int arrSize = 5;
```

```
void main(void)
```

```
{
```

```
    int tests[arrSize] = {87, 75, 98, 100, 82};
```

```
    int result;
```

```
    result = searchList(tests, arrSize, 100);
```

```
    if (result == -1)
```

```
        cout << "You did not earn 100 points on any test\n";
```

```
    else{
```

```
        cout << "You earned 100 points on test ";
```

```
        cout << (result + 1) << endl;
```

```
    }
```

```
}
```

Program Output:

You earned 100 points on test 4.

Discussion

- **Advantage** of linear search is its simplicity
 - Easy to understand
 - Easy to implement
 - Does not require array to be in order (i.e., sorted)
- **Disadvantage** is its efficiency (or complexity)
 - **Worst case** complexity: $f(n) = n+1$
 - Number of steps are proportional to number n of elements in an array
 - If there are 20,000 items in an array
 - Searched data item is stored in the 19,999th element
 - Entire array has to be searched

Binary Search

- Binary search is more efficient than linear search
 - Requires array to be in **sorted order** (i.e., ascending order)

Algorithm works as follows:

- Starts **searching** from the **middle element** of an array
- If value of **data item is less** than the value of middle element
 - Algorithm starts over **searching the first half** of the array
- If value of **data item is greater** than the value of middle element
 - Algorithm starts over **searching the second half** of the array
- Algorithm **continues halving** the array until data item is found

Binary Search Algorithm

```
// numElems - maximum number of elements in the array
// value     - integer data (item) to be searched
// position  - array subscript that holds value (if success)
//           -1 if value not found
int binarySearch(int array[], int numelems, int value)
{
    int first = 0, last = numelems - 1, middle, position = -1;
    bool found = false;
    while (!found && first <= last){
        middle = (first + last) / 2;           // Calculate mid point
        if (array[middle] == value) {         // If value is found at mid
            found = true;
            position = middle;
        }
        else if (array[middle] > value)       //If value is in lower half
            last = middle - 1;
        else
            first = middle + 1;               // If value is in upper half
    }
    return position;
}
```

Binary Search Example

	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]
list	4	8	19	25	34	39	45	48	66	75	89	95

Sorted list for binary search

key = 89

Iteration	first	last	mid	list[mid]
1	0	11	5	39
2	6	11	8	66
3	9	11	10	89

← Value is found

key = 34

Iteration	first	last	mid	list[mid]
1	0	11	5	39
2	0	4	2	19
3	3	4	3	25
4	4	4	4	34

← Value is found

Calling Function binarySearch

```
#include <iostream.h>
// Function prototype
int binarySearch(int [], int, int);
const int arrSize = 20;
```

```
void main(void)
```

```
{
```

```
    int empIDs[arrSize] = {101, 142, 147, 189, 199, 207, 222, 234, 289, 296,
                           310, 319, 388, 394, 417, 429, 447, 521, 536, 600};
```

```
    int searchID = 199;
```

```
}
```

Program Output:

Enter the Employee ID you wish to search for: 199
That ID is found at element 4 in the array

Calling Function binarySearch

```
#include <iostream.h>
// Function prototype
int binarySearch(int [], int, int);
const int arrSize = 20;
```

Program Output:

Enter the Employee ID you wish to search for: 199
That ID is found at element 4 in the array

```
void main(void)
{
    int empIDs[arrSize] = {101, 142, 147, 189, 199, 207, 222, 234, 289, 296,
                           310, 319, 388, 394, 417, 429, 447, 521, 536, 600};

    int result, empID;
    cout << "Enter the Employee ID you wish to search for: ";
    cin >> empID;
    result = binarySearch(empIDs, arrSize, empID);
    if (result == -1)
        cout << "That number does not exist in the array.\n";
    else {
        cout << "That ID is found at element " << result;
        cout << " in the array\n";
    }
}
```

Efficiency Of Binary Search

- Much more efficient than the linear search
- How long does this take (worst case)?
 - If the list has 8 elements
 - It takes 3 steps ($2^3 = 8$)
 - If the list has 16 elements
 - It takes 4 steps ($2^4 = 16$)
 - If the list has 64 elements
 - It takes 6 steps ($2^6 = 64$)
- Worst case complexity: $f(n) = \log_2(n)$
 - Takes $\log_2 n$ steps

Any Question So Far?

